

OE-EC604A — Electronic Measurement & Measuring Instruments

Semester VI • 3L:0T:0P • 3 Credits

Module 1 Exam Notes: Block Schematics & Basic Instruments

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1. Block Schematics of Measuring Systems

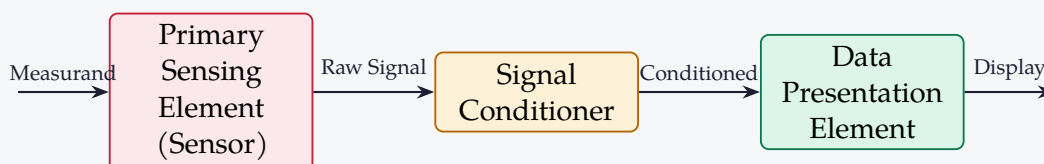
A measuring system is configured to convert a physical quantity (measurand) into an observable visual display or a electrical output.

1.1 General Functional Elements

A standard measuring system consists of three distinct stages:

1. **Primary Sensing Element (Sensor):** Interacts directly with the measurand and converts the physical quantity (e.g., temperature, pressure) into an electrical variable.
2. **Variable Conversion & Signal Conditioning Element:** Amplifies, filters, or modulates the raw sensor output to make it suitable for display or transmission.
3. **Data Presentation Element:** Displays the measured value in an analog pointer deflection or digital form (DSO, LCD, indicator).

Generalized Measuring System Block Schematic



2. Performance Characteristics of Measuring Instruments

Performance parameters of any measuring device are divided into static and dynamic characteristics.

2.1 Static Characteristics

Static characteristics apply when the physical quantity remains constant or changes very slowly.

Key Static Characteristics

- **Accuracy:** The closeness of an instrument reading to the true value of the measurand.
- **Precision:** The degree of reproducibility of measurements (scatter of readings).
- **Resolution:** The smallest change in the input value that can be detected by the instrument.
- **Sensitivity:** The ratio of the change in output deflection ($\Delta\theta_o$) to the change in input physical quantity (Δq_i): $S = \Delta\theta_o / \Delta q_i$.
- **Repeatability:** The closeness of agreement between successive measurements taken under identical conditions over a short time.
- **Reproducibility:** The consistency of readings taken by different observers, locations,

or instruments over a long period.

- **Fidelity:** The ability of a system to reproduce the input signal without dynamic distortion.
- **Static Lag:** The retardation/delay in the response of an instrument to changes in the measurand.

2.2 Types of Measurement Errors

Errors are defined as the difference between the measured value (A_m) and the true value (A_t). They are classified as:

- Gross Errors:** Caused by human errors in reading, recording, or calculating instruments (e.g., parallax errors).
- Systematic (Instrumental/Environmental/Observational) Errors:** Errors inherent in the instrument design, environment (temperature, humidity), or calibration drift. These are predictable and can be corrected.
- Random Errors:** Residual errors that remain after gross and systematic errors have been eliminated. They follow a Gaussian/normal statistical distribution.

2.3 Dynamic Characteristics

Dynamic characteristics apply when the input changes rapidly with time. They include:

- **Speed of Response:** The rapidity with which the instrument responds to changes in the measurand.
- **Measuring Lag:** Delay in response to a dynamic input.
- **Dynamic Error:** The difference between the true value of a quantity and the value indicated by the instrument under dynamic conditions.
- **Fidelity:** True reproduction of dynamic inputs.

3. Measuring Instruments & Range Extension

3.1 D'Arsonval Movement (PMMC)

The **Permanent Magnet Moving Coil (PMMC)** movement is the core analog indicator mechanism. It operates on the electromagnetic force principle: when current flows through a moving coil inside a magnetic field, a deflecting torque is created.

- **Deflecting Torque (T_d):** $T_d = B \cdot A \cdot N \cdot I$ (where B = flux density, A = coil area, N = turns, I = current).
- **Controlling Torque (T_c):** Provided by hair springs: $T_c = K \cdot \theta$ (where K = spring constant, θ = deflection angle).
- **Balance Condition:** At steady state, $T_d = T_c \implies \theta \propto I$ (linear scale).
- **Drawback:** Measures DC only. AC averages to zero.

3.2 Extension of Range

PMMC coils are delicate and can handle only small currents (typically $50\ \mu\text{A}$ to $1\ \text{mA}$). For larger ranges, multipliers and shunts are used:

Range Extension Calculations

1. **DC Ammeter Range Extension (Shunt Resistor R_{sh}):** To measure a total current I using a meter of resistance R_m and full-scale current I_m :

$$I_{sh} \cdot R_{sh} = I_m \cdot R_m \Rightarrow (I - I_m)R_{sh} = I_m R_m \Rightarrow R_{sh} = \frac{R_m}{\frac{I}{I_m} - 1} = \frac{R_m}{m - 1}$$

where $m = I/I_m$ is the ammeter multiplying factor.

2. **DC Voltmeter Range Extension (Multiplier Resistor R_s):** To measure a total voltage V using a meter of resistance R_m and full-scale voltage $V_m = I_m R_m$:

$$V = I_m(R_s + R_m) \Rightarrow R_s + R_m = \frac{V}{I_m} \Rightarrow R_s = R_m \left(\frac{V}{V_m} - 1 \right) = R_m(m - 1)$$

where $m = V/V_m$ is the voltmeter multiplying factor.

3.3 True RMS Responding Voltmeter

Standard AC meters measure the average value and scale it by 1.11 to indicate RMS for perfect sine waves. For non-sinusoidal waves, this causes massive errors. - **Working Principle:** True RMS voltmeters use a **thermocouple/heating element** combination. The input AC voltage is fed into a heater resistor, producing heat proportional to the RMS power ($\text{Heat} \propto V_{\text{rms}}^2$). A thermocouple converts the temperature difference into a DC voltage, driving a standard PMMC meter to indicate the true RMS value of any waveform.

Quick Revision — Module 1

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Key Range Extension Formulas

$$\text{Ammeter Shunt: } R_{sh} = \frac{R_m}{m - 1} \quad \text{where } m = \frac{I}{I_m}$$

$$\text{Voltmeter Multiplier: } R_s = R_m(m - 1) \quad \text{where } m = \frac{V}{V_m}$$

$$\text{Static Error: } E_s = A_m - A_t \quad \text{Static Sensitivity: } S = \frac{\Delta \text{Output}}{\Delta \text{Input}}$$

One-Line Definitions

- **Accuracy:** Degree of closeness of a measured value to the actual true value.
- **Precision:** Degree of agreement/repeatability within a group of measurements.
- **Resolution:** The smallest detectable change in the input measurand.
- **PMMC:** Permanent Magnet Moving Coil movement; sensitive DC indicator based on magnetic coil torque.
- **True RMS Voltmeter:** Instrument measuring heating power of input signal to compute exact RMS regardless of shape.
- **Gross Errors:** Human-induced errors in writing down readings or incorrect usage of instruments.
- **Systematic Errors:** Predictable offset errors from instrument design limits or external environment.

Frequently Asked / Viva Questions

Q1. Differentiate between Accuracy and Precision.

Accuracy is the closeness of a measured value to the true value, whereas precision is the repeatability of readings when measured repeatedly. An instrument can be highly precise (give the same reading constantly) but inaccurate if it has a constant offset error.

Q2. A PMMC meter has internal resistance $R_m = 100 \Omega$ and full-scale deflection current $I_m = 1 \text{ mA}$. Calculate the shunt resistance required to extend its range to 1 A.

Given: $R_m = 100 \Omega$, $I_m = 1 \text{ mA} = 0.001 \text{ A}$, Target current $I = 1 \text{ A}$.

Multiplying factor: $m = I/I_m = 1/0.001 = 1000$.

Shunt resistance: $R_{sh} = R_m/(m - 1) = 100/(1000 - 1) = 100/999 \approx 0.1001 \Omega$.

Frequently Asked / Viva Questions (contd.)

Q3. Using the same PMMC meter ($R_m = 100\ \Omega$, $I_m = 1\ \text{mA}$), calculate the multiplier resistance required to extend its range to measure a voltmeter scale of 100 V.

Full scale meter voltage: $V_m = I_m \cdot R_m = 0.001 \times 100 = 0.1\ \text{V}$.

Multiplying factor: $m = V/V_m = 100/0.1 = 1000$.

Multiplier series resistance: $R_s = R_m(m - 1) = 100 \times 999 = 99,900\ \Omega = 99.9\ \text{k}\Omega$.

Q4. Why does a PMMC instrument fail to measure AC values directly?

Because the direction of deflecting torque depends on the current polarity. For AC, the current alternates polarity rapidly (e.g., 50 Hz). The pointer attempts to follow the oscillations but due to inertia, it stays at zero, displaying only the average value (which is 0 for AC).

Q5. What is True RMS and why is it preferred over average-responding voltmeters?

Average-responding voltmeters rectify AC, find the average, and multiply it by 1.11 (the form factor for a sine wave) to display RMS. This is invalid for square, triangular, or distorted waves. A True RMS voltmeter measures the actual thermal power generated by the signal, representing its true heating equivalent.

Q6. What are systematic errors and how can they be minimized?

Systematic errors are predictable, constant errors arising from instrument calibration, component aging, or environmental changes (temperature/humidity). They can be minimized by periodic calibration, proper shielding, and applying mathematical correction factors.

Q7. Define the term Resolution in measuring instruments.

Resolution is the smallest increment of input quantity change that results in a detectable change in the instrument output/deflection.

Q8. Explain the role of the resident monitor in early batch processing systems.

The resident monitor was the primitive predecessor of the modern OS kernel. It remained permanently in memory, automatically loading, sequencing, and executing sequential jobs from card/tape readers, reducing idle time between jobs.

Q9. What is the difference between Gross Errors and Random Errors?

Gross errors are due to human mistakes like incorrect reading entry or wrong wiring. Random errors are unpredictable, fluctuating variations caused by unknown factors that remain even after all systematic and gross errors have been corrected. They are analyzed statistically.

Q10. What is static lag and how does it differ from dynamic lag?

Static lag is the delay in response when the input measurand is constant or changing very slowly, caused by friction or spring inertia. Dynamic lag occurs when the input changes rapidly, caused by the system's dynamic parameters (mass, damping, capacitance).

Accuracy vs Precision Comparison Summary

Feature	Accuracy	Precision
Core Concept	Closeness to the actual true value.	Closeness of multiple readings to each other.
Error Type	Linked to systematic errors.	Linked to random errors (scatter).
Calibration	Can be improved by calibration.	Calibration does not improve precision.
Dependence	Requires precision to be accurate.	Can be highly precise but inaccurate.